Fiscal Year:	FY 2012	**Task Last Updated:**	FY 04/13/2012
PI Name:	Blancaflor, Elison B. Ph.D.		
Project Title:	Actin Regulation of Arabidopsis Root Growth and Orientation During Spaceflight		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	GOVERNMENT	**Phone:**	321-501-5368
Organization Name:	NASA Kennedy Space Center		
PI Address 1:	Utilization and Life Science Office		
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City:	Kennedy Space Center	**State:**	FL
Zip Code:	32955	**Congressional District:**	8
Comments:	NOTE: Dr. Blancaflor moved to NASA Kennedy Space Center in spring 2021; as a PI, his prior grants were for work at the Noble Foundation in Oklahoma.		
Project Type:	Flight	**Solicitation / Funding Source:**	2009 Space Life Sciences BRIC NNH09ZTT004N
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No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Shuttle/ISS		
Flight Assignment:	STS-131		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Nakashima, Jin (Noble Foundation) Tang, Yuhong (Noble Foundation)		
Grant/Contract No.:	NNX10AF43G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	We conducted experiments utilizing the Biological Research in Canisters (BRIC)- Petri Dish Fixation Unit (PDFU) onboard STS-131 to understand how the actin cytoskeleton dictates root growth orientation during spaceflight. Our guiding hypothesis was that actin filaments (F-actin) negatively regulate environmental and endogenous signals to specify root orientation on earth and in space. This hypothesis was based on previous NASA-funded ground based research (grant NAG2-1518) demonstrating that F-actin disruption enhanced the sensitivity of roots to gravity. We utilized Arabidopsis seedlings with a transfer (T)-DNA mutation in a vegetative actin isoform (act2) and the short-duration microgravity environment provided by the STS-131 mission to test our hypothesis. Microgravity was an essential component of our proposed experiments because it allowed us to determine the impact of other environmental signals on root orientation that are typically masked by the strong gravitational force on earth. Using two unpowered BRIC units, we quantified root orientation of dark-grown act2 and wild-type seedlings (aim 1). In parallel, we conducted an extensive set of genome-wide microarray studies using Affymetrix ATH1 arrays to unravel actin-dependent gene regulatory networks that modulate root growth and orientation during spaceflight (aim 2). Although such experiments required minimal manipulation from the flight crew, we were able to collect a robust dataset that has paved the way for an in depth understanding of plant growth under microgravity, a goal consistent with the overall mission of NASA. Another significant deliverable of our project will be the inclusion of a photo-documentary article that will outline the activities of the selected Arabidopsis BRIC16 proposals in a book that will be published by Gakken Education Publishing Co., Ltd. which is the largest educational publication company in Japan. This will provide NASA with a new venue to highlight the significance of its life sciences space program to the general public. The results of our experiments on STS-131 showed that microgravity triggers the transcriptional reprogramming of genes encoding cell wall-related genes. As a consequence of these changes in gene expression, root growth orientation and root hair tip growth were modified during spaceflight. In addition to providing a deeper understanding to microgravity effects on plant development, results from this spaceflight proposal have uncovered new molecular players that regulate root development.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will lead to a better understanding of how plant development is affected by microgravity. Gravity dictates a number of important plant physiological processes here on Earth that impact agricultural productivity. For example, gravity influences the downward growth of roots and upward growth of shoots (gravitropism), which are critical to optimize nutrient, water, and light acquisition. Through a better understanding of the cellular and molecular processes underlying plant gravitropism, crops with improved growth habits and enhanced architecture can be genetically engineered. Furthermore, understanding root growth behavior in microgravity will guide the development of plant strains better equipped to optimize nutrient and water acquisition, and waste recycling in enclosed regenerative life support systems. Because upright plants evolved under the 1-g conditions of Earth in large part through the mechanical strength of its cell walls, knowledge gained from this proposal will provide a deeper mechanistic appreciation of the genetic circuitry that specifies how cell walls are constructed in microgravity. On Earth, information on cell walls has opened up new strategies to engineer plants with enhanced biomass and reduced recalcitrance for bioenergy research. In space, we expect that such knowledge will enable technologies for designing plants that can sustain optimal growth and stature despite the apparent adverse impact that microgravity has on cell wall development.
Task Progress:	The major goal of this proposal is to determine how the actin cytoskeleton regulates the orientation of root growth and plant development under microgravity conditions. The two aims in support of this goal were: 1). Quantify root orientation and determine amyloplast position in the root columella of Arabidopsis seedlings of wild-type and act2 mutants grown on the Biological Research in Canisters (BRIC)-Petri Dish Fixation Unit (PDFU) and 2). Compare transcript profiles of wild-type and act2 mutants (has a mutation in the vegetative actin 2 isoform) using Affymetrix ATH1 arrays to unravel actin-dependent gene regulatory networks that modulate seedling orientation during spaceflight. We have completed most of the experiments outlined in both aims. Manuscripts for submission to peer-reviewed journals are at various stages of preparation. Results from our spaceflight experiments are summarized below. Fourteen 60 mm Petri dishes containing Arabidopsis seeds were loaded into BRIC-PDFUs and launched to space on the Discovery STS-131 mission. The seeds germinated in space and seedlings grew in the dark for about two weeks. Seedlings were returned to the Space Life Sciences Laboratory (SLSL) at the Kennedy Space Center (KSC) after fixation in orbit by the crew of Discovery. We have completed the microarray studies proposed in aim 2. We found 975 genes to be differentially regulated by spaceflight. In wild-type seedlings for example, 157 genes were elevated by 2-fold or more during spaceflight whereas 779 genes were downregulated. We observed some interesting trends particularly with regard to genes that were downregulated in microgravity. One of the more striking trends was that 20% percent of genes that were downregulated in space were genes associated with plant cell wall function. The number of genes differentially regulated during spaceflight that were predicted or verified to have cell wall-related functions included genes that encoded a large number of class III peroxidases and the super family of hydroxyproline-rich glycoproteins (HRGPs) and their subfamily members such as AGPs, extensins, and proline-rich proteins (PRPs). Interestingly the cell wall genes downregulated in space such as peroxidases and HRGPs were previously reported and/or predicted to be highly elevated in roots and root hairs. Consistent with this observation, functional studies of some of these root hair-expressed genes were reported to have defects in root hair development such as a gene (called COW1) that encodes a sec14p domain phosphatidyl inositol transfer protein. In this regard, we recently discovered that COW1 is involved in coordinating remodeling of the actin cytoskeleton with membrane trafficking to control root hair development. The downregulation of COW1 in space opens the possibility that this gene might be an essential component of pathways that maintain normal signaling between the actin cytoskeleton and secretion of cell wall components during microgravity. Through mutant analysis, we have also discovered genes encoding peroxidases and a lipid transfer protein that regulate root development. We have also made significant progress on the root orientation and structural studies proposed in aim 1. We observed that roots in microgravity exhibited a tendency to skew to the left instead of the random growth that one would expect when gravity is absent. To quantify root orientation from seedlings grown in microgravity, we refined an algorithm that allowed the assignment of a root orientation index to individual root images. This algorithm expanded on software that we had recently developed for the quantification of actin filament organization in roots. On the basis of this analysis, we found that space-grown seedlings had a higher root orientation index, which corresponded to highly misaligned roots (i.e. more intense coiling) whereas the ground controls had a low root orientation index value, which was indicative of a root that grew fairly straight. Interestingly, using this algorithm, we found that roots of act2 mutants exhibited higher root orientation indices compared to wild-type suggesting that the actin cytoskeleton is indeed a major facilitator of the root skewing response in microgravity.

	The downregulation of genes related to cell wall function, the alterations in root hair development, and the skewing of primary roots are indications that changes in root cell ultrastructure might have occurred during spaceflight. We therefore studied ground and space-grown seedlings at the light and transmission electron microscopy (TEM) level. At the light microscope level no dramatic differences in root cellular morphology between ground and space-grown wild-type and act2 mutants were observed. However, at the TEM level, there were clear structural deformities in the cell walls of roots particularly in act2 mutants grown in microgravity. These abnormalities included an increase in the frequency of cell walls that were wavy and cell walls with a significant reduction in electron opaque material. The actin cytoskeleton through myosin-based molecular motors is crucial for the targeted delivery of vesicles carrying cell wall precursors. During spaceflight, the defective actin cytoskeleton in act2 mutants might lead to amplified cell wall abnormalities as shown by the higher incidence of unevenly deposited cell wall material (i.e. cell wall waviness) and lower electron opaque regions in the cell wall. These defects in cell wall formation in the act2 mutants could then translate into more pronounced differential growth between the flanks of the root leading to a stronger coiling response in space. These findings are consistent with our hypothesis that one of gravity's impacts on plant growth is through its influence on the establishment of physical links between the cytoskeleton/plasma membrane and the cell wall to sustain normal root development.
Bibliography Type:	Description: (Last Updated: 01/13/2026)
Abstracts for Journals and Proceedings	Nakashima J, Liao F, Sparks JA, Tang Y, Blancaflor EB. "Ultrastructural and Morphological Studies of Arabidopsis Seedlings in Microgravity." Presented at the 27th Annual Meeting of the American Society for Gravitational and Space Biology and 32nd Annual Meeting of the International Society for Gravitational Physiology, San Jose, California, November 3-6, 2011. Program and abstracts. 27th Annual Meeting of the American Society for Gravitational and Space Biology and 32nd Annual Meeting of the International Society for Gravitational Physiology, San Jose, California, November 3-6, 2011. p. 102. , Nov-2011
Abstracts for Journals and Proceedings	Sparks JA, Tang Y, Nakashima J, Blancaflor EB. "Transcript profiling of space-grown Arabidopsis uncovers a lipid transfer protein that regulates primary root growth." Presented at the 27th Annual Meeting of the American Society for Gravitational and Space Biology and 32nd Annual Meeting of the International Society for Gravitational Physiology, San Jose, California November 3-6, 2011. Program and abstracts. 27th Annual Meeting of the American Society for Gravitational and Space Biology and 32nd Annual Meeting of the International Society for Gravitational Physiology, San Jose, California, November 3-6, 2011. p. 102. , Nov-2011
Abstracts for Journals and Proceedings	Blancaflor EB, Sparks JA, Nakashima J, Tang Y. "Microgravity impacts tip growth in Arabidopsis by down-regulating genes encoding cell wall related proteins." Presented at the 27th Annual Meeting of the American Society for Gravitational and Space Biology and 32nd Annual Meeting of the International Society for Gravitational Physiology, San Jose, California, November 3-6, 2011. Program and abstracts. 27th Annual Meeting of the American Society for Gravitational and Space Biology and 32nd Annual Meeting of the International Society for Gravitational Physiology, San Jose, California, November 3-6, 2011. p. 50. , Nov-2011